L2.3 How Derivatives Affect the Shape of a Graph

Work in your group. Check your answers against the Quarto tonight.

The I/D Test and the sign test

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1. On which intervals is f increasing, and on which is it decreasing?

$$f(x) = x^2 - 6x + 5$$

2. On which intervals is f increasing, and on which is it decreasing?

$$f(x) = 2x^3 - 3x^2 - 12x$$

The First Derivative Test

3. Find the critical numbers of f, then find every local extremum.

$$f(x) = x^{2/3}$$

> **Stop.** f' is never zero here. There is still a critical number — what is the other way to get one?

4. $f(x) = (x-2)^3$ has exactly one critical number. What is it?

Concavity and the Second Derivative Test

5. For $f(x) = x^4 - 4x + 1$: find every PIP, then find the intervals of concavity. For each PIP, say whether it is an inflection point, and **why**.

6. Use the **Second** Derivative Test to classify the critical numbers of

$$f(x) = 2x^3 + 3x^2 - 36x$$

7. (a) Find the local extrema of $f(x) = x^4 - 8x^2$ using **both** the First and the Second Derivative Test. Which one did you prefer, and why?

(b) Now try the Second Derivative Test on $g(x) = (x-1)^4$ at its critical number. What does it tell you? Settle the question another way.

The derivatives table

8. Consider the function

$$\boxed{f(x) = 2x^3 - 9x^2 + 12x - 3}$$

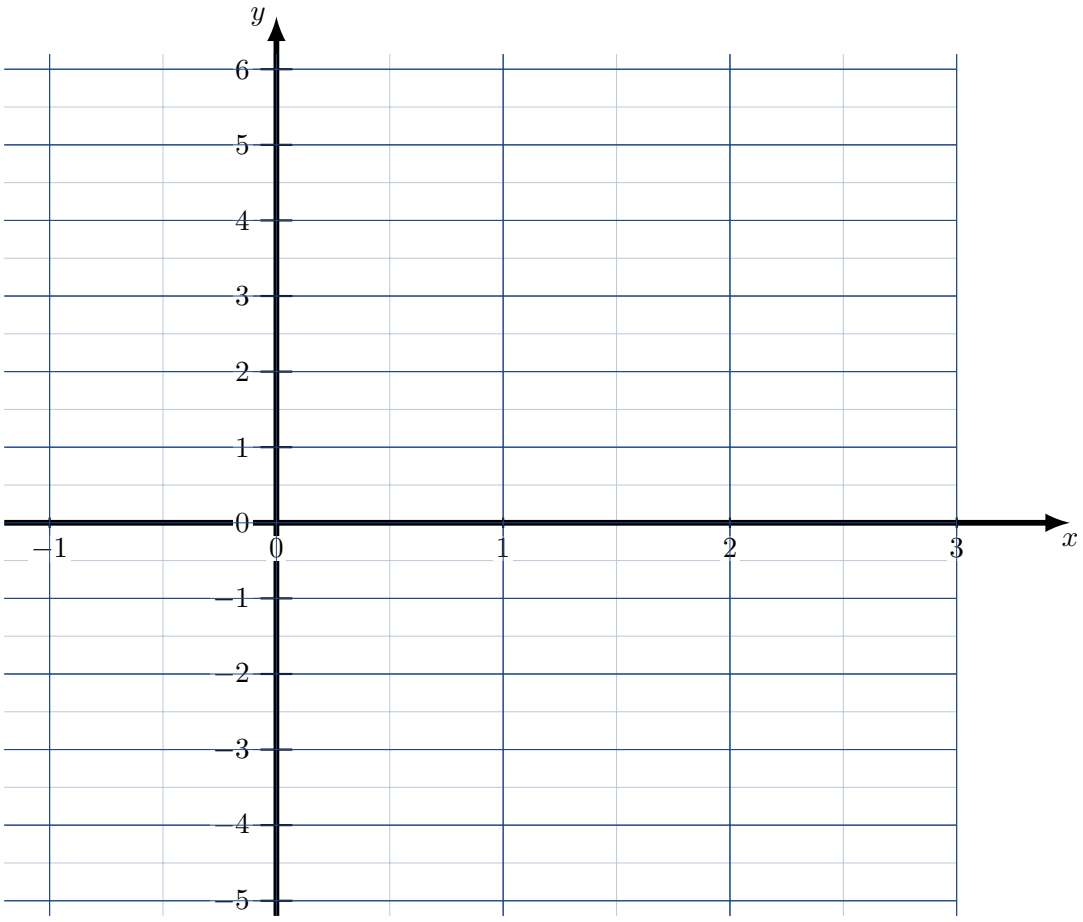
(a) Find the intervals on which f is increasing or decreasing.

(b) Find any local minimum and Maximum values of f.

(c) Find the intervals of concavity and the inflection points.

9. Make the derivatives table for $f(x) = 2x^3 - 9x^2 + 12x - 3$.

10. Sketch $f(x) = 2x^3 - 9x^2 + 12x - 3$ on the axes below. Find the y-intercept. The only x-intercept is ≈ 0.322.



11. Find the CNs, extrema and IPs of

$$f(x) = \frac{2}{x^2 + 1}$$

See if you can sketch it.

Stop. Rewrite it as $2(x^2 + 1)^{-1}$ first.

12. BOARDS **Group work.** A function is defined by

$$\boxed{f(x) = x^3 - 2x^2 - 3x - 1}$$

(a) Graph $f(x)$. Label any inflection points (IP), any local Max points (M) and any local min points (m).

(b) Graph $f'(x)$ on the same axes. What happened to the IPs?

(c) Graph $f''(x)$ on the same axes. What happened to the Ms and ms?

(d) Complete the sentence. IPs become _____ in the graph of the first derivative of a function, which become _____ in the graph of the second derivative.

Repeat (a)–(d) for $\boxed{g(x) = x^3 - 3x - 1}$. How are its graphs different from the graphs of f?